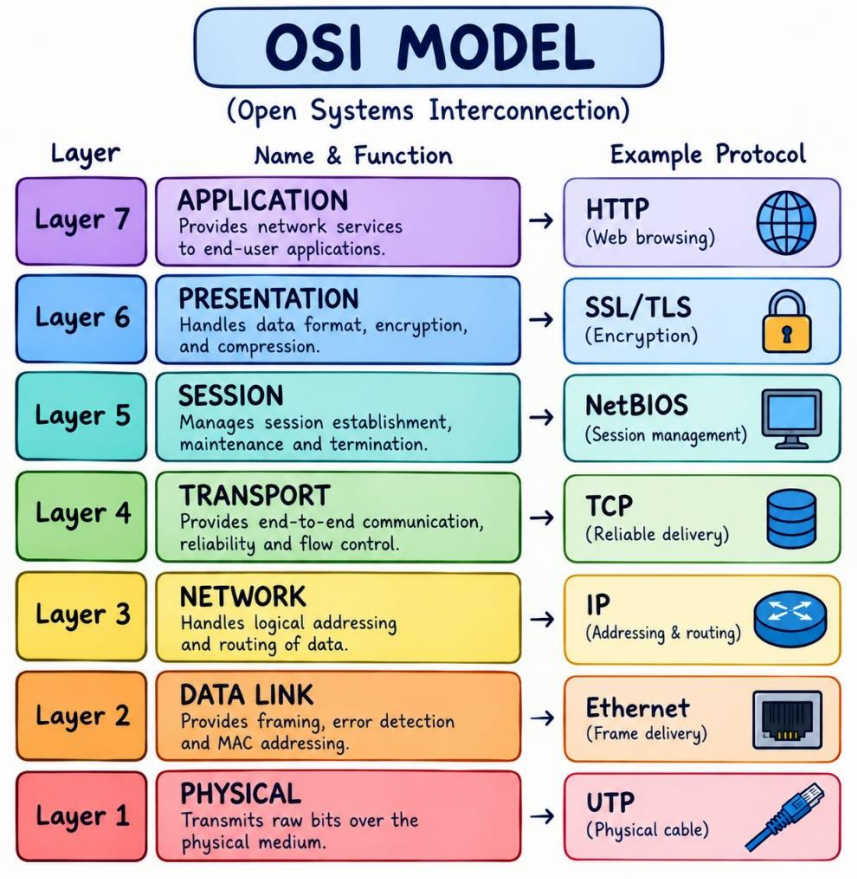
OSI & Network protocols

Tasks

1. Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

Ans.:-



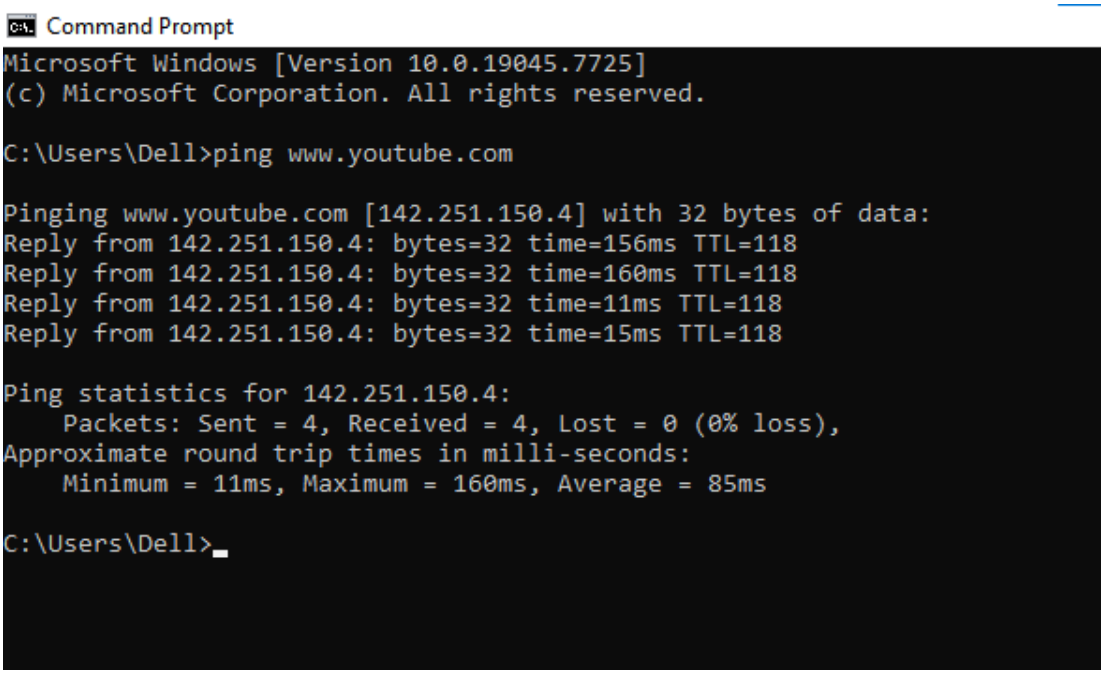
2. Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

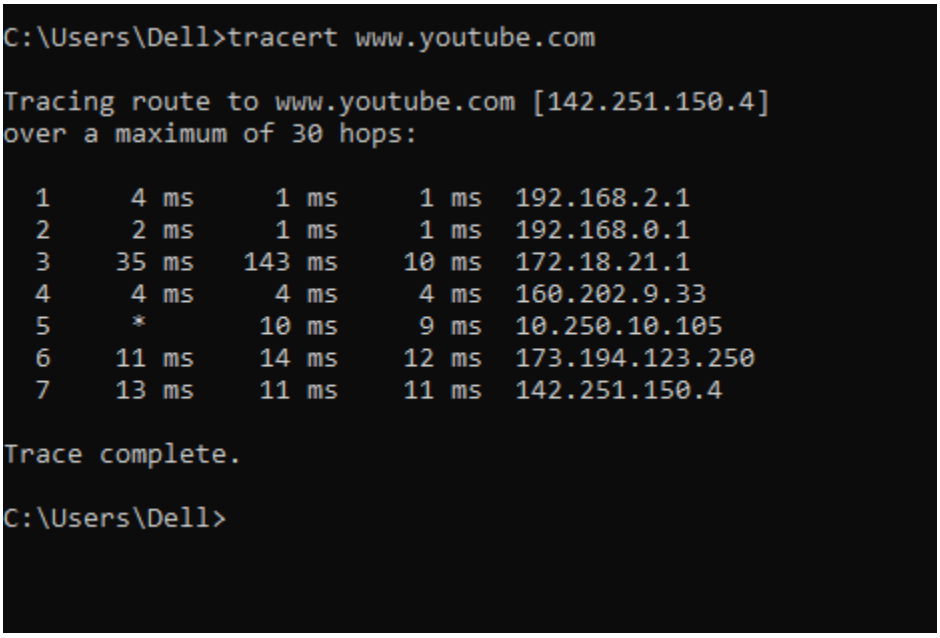
Ans.:-

NO.	APP	PROTOCOL 1	PROTOCOL 2
1	**WhatsApp**	HTTPS	DNS
2	**Instagram**	HTTPS	TCP
3	**Zomato**	HTTPS	DNS
4	**Flipkart**	HTTPS	TCP
5	**Paytm**	HTTPS	DNS

3. Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

Ans.:-





OSI Layers Involved :-

OSI Layer	*Name*	*Role in ping / tracert*
Layer 7	Application	CMD/terminal provides the interface for the commands
Layer 4	Transport	tracert can use ICMP/UDP depending on implementation; ping uses ICMP directly rather than TCP/UDP
Layer 3	Network	IP and ICMP handle addressing, routing, and echo requests/replies
Layer 2	Data Link	Ethernet/Wi-Fi frames carry the packets on the local network
Layer 1	Physical	Electrical/radio signals carry the bits through cable or Wi-Fi

**4. Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step.

Hint: Start from Application layer (your message in WhatsApp) down to Physical layer (actual data sent over WiFi/mobile network), then back up to Application layer on your friend's phone.**

Ans.:-

Suppose I send the message **"Hello, how are you?"** to my friend on WhatsApp. The message travels through all seven OSI layers before reaching my friend's phone.

Layer 7 – Application: I type "Hello, how are you?" in WhatsApp and press Send. WhatsApp prepares the message for transmission.

Layer 6 – Presentation: The data is formatted and encrypted so that it can be transmitted securely.

Layer 5 – Session: A communication session between my WhatsApp device and the required server is maintained while the message is being sent.

Layer 4 – Transport: The message is divided into smaller pieces and transported using suitable transport mechanisms, helping provide reliable communication.

Layer 3 – Network: IP addressing is used to determine where the data needs to go. Routers forward the packets toward the destination.

Layer 2 – Data Link: The network packet is placed into frames and delivered across the local Wi-Fi or mobile network using hardware/MAC addressing.

Layer 1 – Physical: The frames are converted into bits and transmitted as radio signals over Wi-Fi or mobile networks.

When the data reaches my friend's phone, the process happens in reverse. The bits move from Physical → Data Link → Network → Transport → Session → Presentation → Application. Finally, WhatsApp decrypts and displays "Hello, how are you?" on my friend's screen.